

LogStruct: learned prior-anchored gene-graph smoothing for interpretable omics prediction

Draft status: active empirical draft. The code audit, METABRIC 1,000-gene 5-fold run, GTEx 500-gene 5-fold run, Norman 500-gene 5-fold run, and Tabula Sapiens 500-gene donor-held-out run are backed by logged runs under **results/**. The full 5k-10k feature campaign, remaining datasets, ablations, and biological validation are not complete.

Abstract

High-dimensional omics prediction often benefits from biological prior knowledge, but existing network-regularized linear models typically treat the gene graph as fixed. We evaluate LogStruct, a sklearn-compatible PyTorch estimator that learns a Bernoulli-parameterized feature graph anchored to a biological prior, uses the learned graph as an explicit feature-smoothing operator, and retains linear-model interpretability through effective coefficients. This draft currently reports initial METABRIC, GTEx, Norman perturb-seq, and Tabula Sapiens immune-cell experiments. These first results are mixed: LogStruct is competitive with a fixed STRING graph on METABRIC but trails tuned L2 logistic regression; GTEx confirms the pipeline on a high-signal tissue task but again favors logistic regression; on Norman, LogStruct has similar balanced accuracy but higher macro-F1 and top-5 accuracy than the tuned logistic baseline; on Tabula Sapiens, the current run validates the donor-held-out LogStruct pipeline but lacks tuned logistic and celltypist baselines. These results are logged but not yet publication-complete.

1. Introduction

Genomic and single-cell prediction problems often have many more molecular features than independently sampled biological units. Sparse linear models are attractive in this regime because they are simple, reproducible, and inspectable, but ordinary lasso, ridge, and elastic net ignore curated knowledge about gene interactions, pathways, and regulatory structure. Fixed-graph Laplacian methods address this by encouraging neighboring genes to have similar coefficients, but the prior graph is often noisy, incomplete, or mismatched to the phenotype.

LogStruct tests a middle ground. It keeps the estimator linear after a learned feature-smoothing operator, while allowing the gene graph to deviate from a prior through a KL-anchored Bernoulli adjacency. This is designed for two regimes: bulk transcriptomics, where sample size is small and prior structure may improve sample efficiency, and single-cell data, where rare cell types or perturbations have limited support even when the total number of cells is large.

The planned contributions are:

1. A precise implementation audit of LogStruct’s smoothing, Laplacian, and

- KL terms, including effective coefficients for interpretation.
- 2. A controlled empirical comparison against fixed-graph Laplacian models, GELnet-style penalties, classical baselines, gradient boosting, MLPs, and GNNs.
- 3. Biological validation of learned effective coefficients and graph changes using pathway enrichment, marker recovery, gained/lost edges, and regulatory recovery in perturb-seq.

2. Related work

See `paper/related_work.md`. In brief, LogStruct is positioned between fixed-graph sparse linear methods such as Li and Li, Logit-Lapnet, glmgraph, and GELnet, and learned-graph neural methods such as DIAL-GNN. It should be framed as an incremental but testable combination rather than a new modeling paradigm.

3. Method

Let X in $\mathbb{R}^{(n \times p)}$ be a sample-by-feature matrix and let A_0 in $[0,1]^{(p \times p)}$ be a biological prior adjacency. LogStruct learns adjacency logits L and defines:

$$\begin{aligned} A &= \text{zero_diag}(0.5 * (\text{sigmoid}(L / T) + \text{sigmoid}(L / T)^T)) \\ S(A) &= \alpha I + \beta \text{rownorm}(A) \\ Y_{\text{hat}} &= X S(A) W^T + b \end{aligned}$$

The objective for classification is:

$$\begin{aligned} &CE(Y, Y_{\text{hat}}) \\ &+ \lambda_{\text{en}} EN(W) \\ &+ \lambda_{\text{smooth}} \text{trace}(W L_A W^T) \\ &+ \lambda_{\text{kl}} KL(q(A) || p(A_0)) \end{aligned}$$

Regression replaces cross-entropy with mean squared error. $EN(W)$ is the elastic-net penalty. $L_A = D_A - A$ is the graph Laplacian. The KL term is a Bernoulli KL over adjacency probabilities. This branch adds an `offdiag_mean` KL reduction to avoid $O(p^2)$ scaling of the penalty.

For interpretation, the raw head weights W are not the coefficients on the original feature space. The effective coefficients are:

$$W_{\text{effective}} = W S(A)^T$$

All biological interpretation in this paper must use `effective_coef_`, not `raw_coef_`.

4. Experimental setup

Planned datasets:

- METABRIC PAM50 breast cancer subtype classification.
- TCGA pan-cancer classification, using the UCI subset for fast shakedown and UCSC Xena for the full 33-class run.
- GDSC drug-response regression.
- GTEx broad tissue classification.
- Tabula Sapiens immune-compartment cell-type classification with donor-held-out splits.
- Norman 2019 perturb-seq perturbation identity classification and regulatory recovery.

All experiments will use fixed train/validation/test splits, train-only preprocessing, bootstrap confidence intervals, paired significance tests, and per-fold prediction logs. No result enters this manuscript unless it appears in a `results.json` file with seed, hyperparameters, git SHA, wall-clock time, and hardware record.

5. Results

5.1 Code audit and smoke test

The code audit identified and fixed three implementation issues: missing effective coefficients, temperature-distorted prior probabilities in the KL anchor, and missing STRING alias resolution. It also added unthresholded `adjacency_raw_` to distinguish the prediction-time graph from the thresholded interpretation graph.

The synthetic smoke test in `results/smoke/results.json` passed with balanced accuracy 0.773, macro-F1 0.777, and OvR AUROC 0.965. This verifies the training pipeline only and is not a biological result.

5.2 Initial METABRIC comparison

The first real-data run uses cBioPortal METABRIC mRNA z-scores with CLAUDIN_SUBTYPE labels, keeping LumA, LumB, Her2, Basal, and Normal samples and excluding claudin-low, missing, and NC labels. The processed matrix contains 1,756 samples and 20,385 unique gene symbols. Each outer fold selects the top 1,000 genes by variance on the train/validation partition only. Results are logged in `results/metabric_pam50/results.json`; the table is regenerated from that file by `experiments/01_brca_pam50/make_outputs.py`.

Table: `paper/tables/table1_metabric_1000g.md`.

Key result: tuned L2 logistic regression is best in this first run with balanced accuracy 0.696 (95% bootstrap CI 0.674, 0.715). LogStruct STRING-700 and the frozen fixed-graph STRING-700 baseline are nearly tied at 0.670 balanced accuracy. This does not support a performance advantage for learned adjacency on this reduced METABRIC setting. It does show that the graph-smoothed linear model is competitive but currently weaker than a strong classical baseline.

5.3 Prior sensitivity

Pending real-data runs.

5.4 Frozen versus learned adjacency

Pending real-data runs.

5.5 Smoothing-operator ablation

Pending real-data runs.

5.6 GTEx tissue sanity check

GTEx v8 gene TPM and sample annotations were downloaded from the GTEx Google Storage release. The first sanity-check run uses broad **SMTS** tissue labels, $\log_2(\text{TPM} + 1)$ expression, and 500 genes selected by variance on the training partition of each fold.

Table: `paper/tables/table_gtex_500g.md`.

As expected for tissue-of-origin prediction, tuned L2 logistic regression was near ceiling with balanced accuracy 0.979 (95% CI 0.966, 0.990) and macro-F1 0.974. LogStruct identity and STRING-700 were lower, with balanced accuracy 0.886 and 0.881 respectively. This validates that the data-processing path is not broken, but it does not support using LogStruct over a simple tuned linear baseline for high-signal, high-sample tissue classification.

5.7 Single-cell cell-type classification

Tabula Sapiens v2 Blood, Spleen, and Lymph Node H5AD files were downloaded from CELLxGENE. The custom single-cell runner keeps the full 60,606-gene sparse matrix until each fold selects 500 genes by variance on training donors only. For this first tractable run, classes with fewer than 500 cells were removed and each remaining class was capped at 3,000 cells, yielding a documented 19-class donor-held-out task. This cap makes the result a controlled first experiment, not the final full-atlas claim.

Table: `paper/tables/table_tabula_500g.md`.

Across five donor-held-out folds, LogStruct identity reached balanced accuracy 0.672 (95% bootstrap CI 0.644, 0.704) and macro-F1 0.650. LogStruct STRING-700 was similar at balanced accuracy 0.668 and macro-F1 0.646. The near tie between identity and STRING priors suggests that the current setting is dominated by the learned smoothing/linear classifier rather than the PPI prior. A tuned logistic baseline, celltypist, rare-class bins, and marker recall remain required before making a single-cell performance claim.

5.8 Perturbation identity and regulatory recovery

Norman 2019 data were downloaded from Figshare article 24688110 and loaded from the labeled H5AD. This file is a prefiltered 27,658-cell by 2,000-gene object. For the closed-set run, double perturbations were excluded and controls plus single-gene perturbations with at least 50 cells were retained, yielding 15,216 cells, 18 classes, and 2,000 genes. Five outer folds were run with the top 500 train-selected genes.

Table: `paper/tables/table_norman_500g.md`.

Tuned L2 logistic regression had the highest balanced accuracy at 0.560 (95% CI 0.542, 0.579), narrowly ahead of LogStruct identity at 0.559 and LogStruct STRING-700 at 0.558. However, the LogStruct models were clearly better on macro-F1 and top-5 accuracy: STRING-700 reached macro-F1 0.599 and top-5 accuracy 0.944, compared with logistic macro-F1 0.478 and top-5 accuracy 0.928. This supports a more modest claim than originally hypothesized: the smoothed linear model improves imbalanced perturbation-class ranking and macro-F1 at this feature scale, but not balanced accuracy.

5.9 Biological interpretation

Pending real-data runs.

6. Discussion

The audit already clarifies two important limitations. First, dense adjacency learning materializes $\mathbf{p} \times \mathbf{p}$ matrices and therefore has an $\mathcal{O}(p^2)$ memory ceiling. Second, KL scaling is not a minor numerical detail: the legacy summed KL can dominate at large feature counts, so the paper must report the chosen KL normalization and sensitivity to `lambda_kl`.

The first METABRIC result is a useful negative control for the paper’s honesty standard: LogStruct did not beat a tuned L2 logistic baseline, and the learned STRING graph barely differed from the frozen graph baseline at the tested settings. The likely explanations are the reduced 1,000-gene feature set, short early-stopped neural optimization, and a KL/smoothness setting that may not move the adjacency enough. The next experiment should sweep `lambda_kl`, `lambda_smooth`, and temperature before making any claim about adaptive graph benefit.

The Norman result points in the opposite direction: with imbalanced perturbation classes, the smoothed linear model improved macro-F1 and top-5 accuracy over tuned logistic regression while essentially tying balanced accuracy. This should still be tested with DoRothEA priors before it becomes a biological-regulatory claim.

7. Conclusion

Pending real-data runs.

Availability

Code branch: `experiments/v1`. A release, Zenodo DOI, and reproducible Table 1 notebook remain pending.